No Public Raw File: A Provenance Audit of the Artemis II Lunar Flyby Photograph

The Artemis II flyby photo looks genuine by every internal check, but the raw NEF files named in its own metadata are not publicly available - so the claim stays provisional, and provenance is itself the reproducibility question.

ABSTRACT

We perform a primary-source provenance audit on a single high-stakes empirical object: the Artemis II crewed lunar-flyby photograph art002e012673 (Nikon D5, 400mm, 1/500s f/5.6 ISO 400, no crop, captured 2026-04-06 23:02:32 UTC), published by NASA. Three independent internal checks support the claim that the image shows the lunar surface from a real flyby: (1) exposure/EXIF metadata is internally consistent; (2) a field-of-view and scale analysis yields ~106 m/px ground sampling at the flyby range, consistent with the published trajectory and excluding an Earth-based telescopic capture, which is physically capped near ~1-2 km/px at this focal length; and (3) Gabor texture statistics correlate with Apollo lunar-surface frames at 0.81-0.89. Cross-registration against the LRO WAC 100 m/px global mosaic, however, failed to reproduce a stable match, and - critically - the primary source named in the image's own XMP metadata (NEF raw files art002e012673.NEF and 015B0790.NEF) could not be located in any public archive. We argue that a processed image published without its underlying raw file leaves even a well-supported authenticity claim provisional. The audit doubles as a reproducibility argument: for publicly-funded missions, the raw capture is the reproducibility artifact, and its absence is measurable.

HYPOTHESIS

A high-stakes empirical image claim (a crewed lunar flyby) can only be settled to the level of independent verification through its primary RAW source; a processed derivative, no matter how internally consistent, remains a provisional claim. We test the claim that the public archive contains the raw capture, and treat the absence of that raw file as a quantifiable reproducibility gap rather than a conclusion about the image itself.

EXPERIMENT PLAN

1. Retrieve official metadata from the NASA images-assets asset endpoint for art002e012673 (lens, camera serial, exposure, crop flags, XMP preserved file names). 2. Compute ground sampling from focal length (400mm), sensor size (36mm), and range (~6545 km), compare against the ~100 m/px LRO reference. 3. SIFT cross-register the image against a 7x7 grid of LRO WAC crops with CLAHE and null-test controls (self-match and impossibility controls). 4. Compare texture statistics (tone, Gabor correlation) against Apollo 16/17 surface frames. 5. Search PDS, images-assets, and ASU archives for the named NEF raw files.

CONCLUSION

Internal consistency of the processed image is strong: exposure, FOV/scale, and texture all favor a genuine flyby capture, and the Earth-based-telescope hypothesis is excluded on optics alone. But the definitive primary source - the Nikon NEF raw files explicitly named in the image's XMP metadata - is not located in any public archive, and cross-registration to the reference lunar map is inconclusive. We conclude that the photograph's authenticity is well-supported but not independently settled, and that 'no raw file' is a concrete, measurable reproducibility gap that a provenance-first scientific culture should require to be closed.

CLAIMS

1. The photograph's internal metadata (Nikon D5, 400mm, 1/500s f/5.6 ISO 400, no crop) is internally consistent with a real crewed lunar flyby, not an Earth-based telescope capture.
2. A FOV/scale analysis yields ~106 m/px ground sampling, consistent with the published flyby trajectory and excluding a telescopic capture, which is capped near 1-2 km/px at this focal length.
3. Gabor texture statistics correlate with genuine Apollo lunar-surface frames at 0.81-0.89, supporting a real lunar surface.
4. The primary source - the NEF raw files art002e012673.NEF and 015B0790.NEF named in the image's own XMP metadata - could not be located in any public archive.
5. For a publicly-funded mission, the raw capture is the reproducibility artifact; its absence leaves even a well-supported authenticity claim provisional.

LIMITATIONS

1. Color comparison against Apollo frames was impossible because the archived Apollo surface frames are monochrome.
2. Cross-registration matches the reference at nearly equal resolution (~106 vs 100 m/px), so the null result may be an oblique-view and illumination methodological limit rather than an anomaly.

REFERENCES

1. NASA images-assets metadata for art002e012673 - NASA, 2026
https://images-assets.nasa.gov/asset/art002e012673

2. LRO LROC WAC Global Morphologic Mosaic (100 m/px) - USGS Astropedia, 2013
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